

## LAB 1: INSPECTING TABLE SIZE

### Check table vs index size

```
1 SELECT
2     c.relname AS object_name,
3     pg_size_pretty(pg_table_size(c.oid)) AS data_size,
4     pg_size_pretty(pg_indexes_size(c.oid)) AS index_size,
5     pg_size_pretty(pg_total_relation_size(c.oid)) AS total_size
6 FROM pg_class c
7 JOIN pg_namespace n
8     ON n.oid = c.relnamespace
9 WHERE c.relkind = 'r'
10    AND n.nspname = 'public'
11 ORDER BY pg_total_relation_size(c.oid) DESC;
```

|    | object_name<br>name | data_size<br>text | index_size<br>text | total_size<br>text |
|----|---------------------|-------------------|--------------------|--------------------|
| 1  | payment             | 896 kB            | 1488 kB            | 2384 kB            |
| 2  | rental              | 1232 kB           | 1120 kB            | 2352 kB            |
| 3  | film                | 736 kB            | 216 kB             | 952 kB             |
| 4  | film_actor          | 272 kB            | 216 kB             | 488 kB             |
| 5  | inventory           | 232 kB            | 208 kB             | 440 kB             |
| 6  | customer            | 96 kB             | 112 kB             | 208 kB             |
| 7  | address             | 88 kB             | 64 kB              | 152 kB             |
| 8  | film_category       | 72 kB             | 40 kB              | 112 kB             |
| 9  | city                | 64 kB             | 48 kB              | 112 kB             |
| 10 | actor               | 40 kB             | 32 kB              | 72 kB              |
| 11 | store               | 8192 bytes        | 32 kB              | 40 kB              |
| 12 | staff               | 16 kB             | 16 kB              | 32 kB              |
| 13 | language            | 8192 bytes        | 16 kB              | 24 kB              |
| 14 | country             | 8192 bytes        | 16 kB              | 24 kB              |
| 15 | category            | 8192 bytes        | 16 kB              | 24 kB              |
| 16 | rental_log          | 8192 bytes        | 8192 bytes         | 16 kB              |

The index-heavy tables are the tables whose index size is relatively large compared to their data size: payment, customer

## LAB 2: THE IMPACT OF A NEW INDEX

### 1: Current size of payment indexes

```
1 SELECT pg_size_pretty(pg_indexes_size('payment'));
```

| Data Output                                                                        |                        | Messages | Notifications |
|------------------------------------------------------------------------------------|------------------------|----------|---------------|
|  |                        |          |               |
|                                                                                    | pg_size_pretty<br>text |          |               |
| 1                                                                                  | 920 kB                 |          |               |

Before creating the new index: 920 kB

### 2: Create a heavy index on payment\_date

```
1 CREATE INDEX idx_payment_date_long
2 ON payment (payment_date, amount, customer_id);
```

| Data Output                             |  | Messages | Notifications |
|-----------------------------------------|--|----------|---------------|
| CREATE INDEX                            |  |          |               |
| Query returned successfully in 54 msec. |  |          |               |

Index created successfully.

### 3: Check size again

```
1 SELECT pg_size_pretty(pg_indexes_size('payment'));
```

| Data Output                                                                          |                        | Messages | Notifications |
|--------------------------------------------------------------------------------------|------------------------|----------|---------------|
|  |                        |          |               |
|                                                                                      | pg_size_pretty<br>text |          |               |
| 1                                                                                    | 1488 kB                |          |               |

After creating the new index: 1488 kB

-> The new index consumed 71 pages, where each PostgreSQL page is 8 KB.

## LAB 3: PARTIAL INDEX PRACTICE

```
1 CREATE INDEX idx_expensive_films
2 ON film (replacement_cost)
3 WHERE replacement_cost > 25.00;
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 51 msec.

Partial index created successfully.

```
1 SELECT pg_size_pretty(pg_relation_size('idx_expensive_films'))
```

Data Output Messages Notifications

Showing rows: 1 to 1

|   | pg_size_pretty<br>text |
|---|------------------------|
| 1 | 16 kB                  |

Index size: 16kB

```
1 CREATE INDEX idx_full_replacement_cost
2 ON film (replacement_cost);
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 44 msec.

```
1 SELECT
2     indexrelname,
3     pg_size_pretty(pg_relation_size(indexrelid)) AS index_size
4 FROM pg_stat_user_indexes
5 WHERE indexrelname IN (
6     'idx_expensive_films',
7     'idx_full_replacement_cost');
```

Data Output Messages Notifications

Showing rows: 1 to 2

|   | indexrelname<br>name      | index_size<br>text |
|---|---------------------------|--------------------|
| 1 | idx_expensive_films       | 16 kB              |
| 2 | idx_full_replacement_c... | 40 kB              |

The partial index is 16 kB, compared to 40 kB for the full index. It saves 24 kB of storage (60%) because it only indexes films with `replacement_cost > 25.00`.

Therefore, partial indexes are useful for queries that frequently access a specific subset of data.